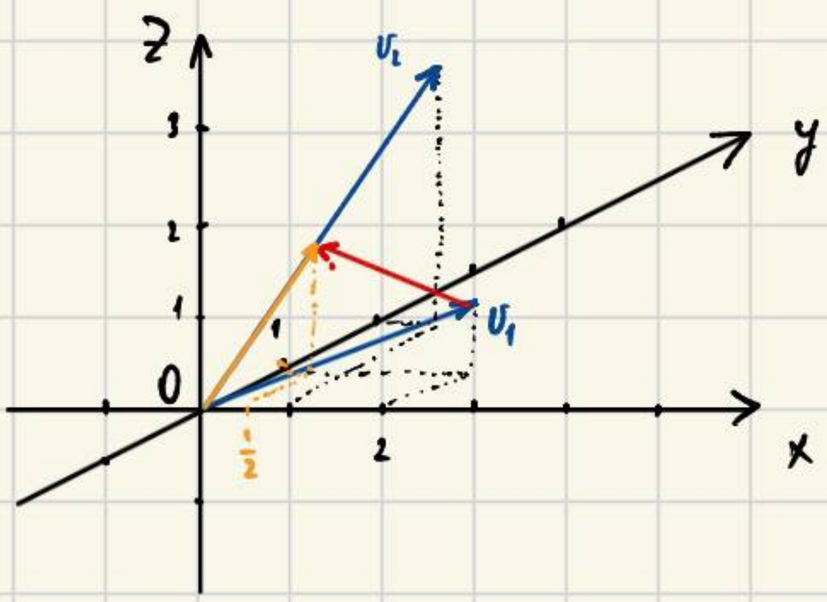
Math Refresher for Masters

Homework 2.

1 $v_1 = (2, 1, 1)$ $\quad v_2 = (1, 2, 3)$

1) $\text{proj}_{v_2}(v_1) = \dfrac{\langle v_1, v_2 \rangle}{\|v_2\|} v_2 = \dfrac{2\cdot1 + 1\cdot2 + 1\cdot3}{1\cdot1 + 2\cdot2 + 3\cdot3}(1,2,3) = \dfrac{7}{14}(1,2,3) = \boxed{\left(\dfrac{1}{2}, 1, \dfrac{3}{2}\right)}$

2)



2 P_2. $\quad p(x) = ax^2 + bx + c$ $\quad p_1 = 1$ $\quad p_2 = x$ $\quad p_3 \neq 0$

$\langle u, v \rangle = \int_{-1}^{1} u(x) \cdot v(x)\, dx$ $\quad \langle p_3, p_1 \rangle = 0$ $\quad \langle p_3, p_2 \rangle = 0$

$p_3(x) = ax^2 + bx + c$

1) $\langle p_3, 1 \rangle = 0 \quad \Rightarrow \int_{-1}^{1} (ax^2 + bx + c)\, dx = 0$

$$\left(a\frac{x^3}{3} + b\frac{x^2}{2} + cx\right)\Big|_{-1}^{1} = 0$$

$$\left(\frac{a}{3} + \frac{b}{2} + c\right) - \left(-\frac{a}{3} + \frac{b}{2} - c\right) = 0$$

$$\frac{2}{3}a + 2c = 0$$

$$a = -3c$$

2) $\langle p_3, x \rangle = 0 \quad \Rightarrow \int_{-1}^{1} (ax^2 + bx + c)x\, dx = 0$

$$\left(a\frac{x^4}{4} + b\frac{x^3}{3} + c\frac{x^2}{2}\right)\Big|_{-1}^{1} = 0$$

$$\left(\frac{a}{4}+\frac{b}{3}+\frac{c}{2}\right)-\left(\frac{a}{4}-\frac{b}{3}+\frac{c}{2}\right)=0$$

$$\frac{2}{3}b=0 \quad \Rightarrow \quad b=0$$

$$p_3(x)=ax^2+bx+c=-3cx^2+c$$

if $c=1$, then $\boxed{p_3(x)=-3x^2+1}$

3 $H=\{x\in\mathbb{R}^3 : \langle w,x\rangle+b=0\}$

$w=(1,1,1) \qquad b=2$

$x=(x,y,z) \quad \Rightarrow \quad \langle w,x\rangle=x+y+z \quad \Rightarrow \quad x+y+z+2=0$

$x+y+z=-2$

$x_1, x_2\in H$, for example: $x_1=(-2,0,0)$

$x_2=(-1,-1,0)$

$$v=x_2-x_1=(-1,-1,0)-(-2,0,0)=\boxed{(1,-1,0)}$$

$$\cos\theta=\frac{\langle v,w\rangle}{\|v\|\cdot\|w\|}=\frac{1\cdot1-1\cdot1+1\cdot0}{\|v\|\cdot\|w\|}=0 \quad \Rightarrow \cos\theta=0 \quad \Rightarrow \boxed{\theta=\frac{\pi}{2}}$$